MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 Admission Onwards)

MDC Course- MG1MDCBCH100- Sports Biochemistry: The Science of Exercise and Human Performance

Duration: 1 Hour Maximum Marks: 35

Students should attempt at least one question from each outcome to enhance their overall outcome attainability

Part A
Multiple Choice Questions
(Answer all questions; each question carries 1 mark)

1. What is the primary focus of Sports Biochemistry? [K] [1]
   a) Energy and Hormonal Metabolism b) Muscle Biochemistry c) Nutritional Biochemistry d) All of the three
2. Which of the following mineral promote the building of strong bones [U] [2]
   a) Calcium b) Zinc c) Iron d) Iodine
3. Which macronutrient is primarily responsible for providing energy during high-intensity athletic activities? [U] [3]
   a) Fat b) Protein c) Fiber d) Carbohydrate
4. Which of the following is a key enzyme released during muscle damage? [U] [5]
   a) Troponin b) C-reactive protein c) Creatine Kinase d) Myoglobin
5. Which of the following is considered an ergogenic aid that is often used to enhance athletic performance? [K] [4]
   a) Creatine b) Vitmain C c) Water d) Caffeine free soda

(1 x 5= 5)

Part B
Fill in the Banks
(Answer all questions; each question carries 2 marks)

6. Cryotherapy involves applying …….to injured areas to reduce inflammation and pain [U] [5]

7. …….. protects from damage caused by free radicals during exercise [U] [1]

8. ……is the function of cardiac muscle [U] [3]

9. ……is a macronutrient necessary for building and repairing of tissues [U] [2]

10. ……is a nutritional did that delays muscle fatigue [U] [4]

(2 x 5=10)

Part C
Short Answer Questions
(Answer any 5 questions; each question carries 3 marks)

11. Explain hormonal adaptations in muscle during exercise [E] [3]
12. Define dehydration. Explain the physiology responses to dehydration during exercise [A] [2]
13. Explain the role of caffeine in the sports performance and endurance [Ap] [4]
14. What is the role of post exercise nutrition in recovery from sports injury? [An] [3]
15. What role does nutrition play in the recovery process after a sports injury? [A] [5]
16. Explain the role of biochemistry in sports science and medicine [A] [1]
17. Identify the advantages of thermotherapy [E] [5]

(3 x 5=15)

Part D
Short Essay Type Question
(Answer any one question; each question carries 5 marks)

18. Demonstrate the mechanism of muscle contraction [An] [3]
19. Discuss the role of nutrition in the recovery process from sports injuries. [A] [5]

(1 x 5 = 5)